Venkata Sai

Software Development Engineer

venkatasai.jobs@gmail.com | +1(512)-848-0295 | Austin, TX

Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Persistent Systems | Identity & Access Management Platform

- Architected and implemented critical components for an **AWS Cognito**-like **identity management** system using **Java** and **Spring Boot**, supporting over 500,000 active user authentications daily.
- Developed highly **scalable authorization** services with **multi-threaded** processing on **AWS Lambda**, handling 10,000 requests per second and reducing latency by 15%.
- Designed and deployed a **distributed system** for secure user data replication across **AWS** regions utilizing **Apache Kafka**, ensuring 99.99% data consistency and availability.
- Led the integration of third-party **federation** protocols like **SAML** and **OAuth 2.0** with existing backend services, expanding partner connections by 25%.
- Implemented advanced **security** features and robust data encryption using **AWS KMS** for sensitive user information, achieving stringent compliance with industry standards.
- Optimized **backend services** using **Java** and **DynamoDB** to efficiently manage user profiles, improving database query performance by 30% for millions of records.

Software Engineer *Jul 2022 – Jun 2025*

Nagarro | Enterprise Backend Services

- Developed and maintained **RESTful APIs** using **Python** and **Flask** for an enterprise application, processing over 1 million requests daily with 99.9% uptime.
- Collaborated on architecting a fault-tolerant message queuing system using **Apache Kafka** for asynchronous processing, improving data ingestion rates by 35% and system reliability.
- Contributed to the design of a **multi-tiered authentication** system, implementing **JWT**-based **security** tokens for improved user access control by 20%.
- Integrated **AWS S3** for secure document storage and retrieval, managing terabytes of client data efficiently while reducing overall storage costs by 18%.
- Implemented **CI/CD pipelines** with **Jenkins** and **Docker** for automated testing and rapid deployment, accelerating release cycles by 40% for new features.
- Performed extensive **code reviews** and upheld stringent **coding standards** for critical **backend services**, reducing production bugs by 25% across releases.
- Managed and optimized **PostgreSQL** databases, including schema design and query tuning, enhancing application response times by 15% for complex operations.
- Developed custom monitoring tools with **Prometheus** and **Grafana** for comprehensive system health and performance, proactively identifying issues 30 minutes faster.

Software Engineer *Jul 2020 – Aug 2022*

Endava | Financial Services Platform

- Developed and maintained core modules for a financial services platform using **Java** and **Spring Framework**, supporting daily transactions for over 10,000 users.
- Implemented robust **RESTful APIs** for client-facing features, improving data exchange efficiency by 20% and significantly reducing integration efforts for partners.
- Utilized **SQL Server** for secure transaction data management, optimizing complex queries to effectively reduce report generation time by 30%.
- Contributed to a significant migration project from monolithic architecture to **microservices**, improving system modularity and deployment flexibility by 15%.
- Participated in the full **software development life cycle**, including requirements gathering, design, comprehensive testing, and deployment, ensuring high-quality software delivery.
- Developed automated unit and integration tests using **JUnit** and **Mockito**, increasing code coverage by 25% and significantly reducing post-release regression defects.

Skills

Languages: Java, Python
Frameworks & Tools: Spring Boot, Spring Framework, Flask, Apache Kafka, Spark Streaming, Jenkins, Docker, Prometheus, Grafana, JUnit, Mockito
Cloud & Infra: AWS, AWS Cognito, AWS Lambda, AWS KMS, AWS S3, SAML, OAuth 2.0
Databases & Search: DynamoDB, PostgreSQL, SQL Server
Concepts: Identity Management, Authorization, Authentication, Federation, Security, Scalable Architectures, Multi-threaded, Distributed Systems, Object-Oriented Design, RESTful APIs, CI/CD, Code Reviews, Coding Standards, Multi-tiered Architecture, Microservices, Software Development Life Cycle, Testing, JWT

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India